MILITARY SPECIFICATION

AIR CONDITIONER A/E32C-25

1. SCOPE

1.1 This specification covers the requirements for a 400-Hz, electric-motor-driven military, compact, vertical, lightweight air conditioner, designated A/E32C-25, having a nominal cooling capacity of 36,000 BTU/hr and a nominal heating capacity of 28,600 BTU/hr.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents form a part of this specification to the extent specified herein. The applicable issues of the governmental documents shall be the issues in effect on date of invitation for bids or request for proposal. The applicable issue of the nongovernmental document shall be the issue available in the files of the procuring activity and made a part of the invitation for bids or request for proposal.

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood

Military

MIL-P-116	Preservation, Methods of
MIL-A-38339	Air Conditioner, Lightweight, Compact, General Requirements for
MIL-D-38386	Duct Assembly, Ground, Conditioned Air, Insulated, Flexible
MIL-C-38392	Compressor, Hermetic, Air Conditioning, General Specification for
MIL-S-52059	Shelter, Electrical Equipment S-141( )/G
MIL-S-55286	Shelter, Electrical Equipment S-280( )/G

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

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DRAWING

Trane Company

DL A4524-5450-2 Model A/E32C-25 Air Conditioner, Trane Model MAC 40 Reprocurement Data Package

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The requirements of MIL-A-38339 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

* 3.1.1 Drawings. The requirements of Trane Company Drawing list DL A4524-5450-2 and detail drawings thereto apply as requirements of this specification with the exceptions and additions specified herein (see 6.3).

3.2 Design and construction

3.2.1 Electric power source. The air conditioner shall be designed to operate from an alternating-current power source with the following characteristics:

a.	Potential (steady)	208V +10 -5 percent
b.	Frequency	400 Hz ±5 percent
c.	Phase	3-phase, 4-wire.

3.3 Performance

3.3.1 Cooling capacity. The preproduction units shall have a minimum cooling capacity of 38,000 BTU/hr and the production units a minimum cooling capacity of 36,000 BTU/hr when operating under the rated conditions specified in MIL-A-38339. At any rated capacity condition, the total power draw shall not exceed 12.4 kw.

3.3.2 Heater capacity. The power consumption of the electric resistance heater shall be 8,400w, ±5 percent, exclusive of the fan motor.

* 3.4 Evaporator assembly. The evaporator assembly shall be physically interchangeable (in the air conditioner) with the evaporator assembly depicted on the drawings (see 3.1.1) and shall meet the performance requirements specified herein. If required, in order to meet the dehumidification requirements and tests of MIL-A-38339, a mist eliminator shall be included as part of the evaporator assembly.

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3.4.1 Evaporator airflow. When the air conditioner is operating in 125° F ambient and with 90° F dry-bulb and 75° F wet-bulb return air to the evaporator, the evaporator fan shall provide a minimum of 1,375 cfm at free-air delivery and a minimum of 1,080 cfm against a static pressure of 1.0 inch of water.

* 3.5 Condenser assembly. The condenser assembly shall be physically interchangeable (in the air conditioner) with the condenser assembly depicted on the drawings (see 3.1.1) and shall meet the performance requirements specified herein.

3.6 Compressor. The compressor shall meet the performance requirements of MIL-C-38392 as specified for class 4, type II compressors.

3.7 External flexible ducts. When specified (see 6.2), two 12-inch-diameter external flexible ducts in accordance with MIL-D-38386 shall be furnished with each air conditioner.

3.7.1 Duct adapter. When ducts are required, an adapter shall be furnished to provide a transition between the air conditioner evaporator supply and return air openings and the ducts. The adapter shall conform to the drawing specified in MIL-A-38339.

* 3.8 Canvas cover. When specified (see 6.2), an external canvas cover shall be furnished in accordance with the drawings (see 3.1.1).

3.9 Noise level. The noise level generated by the air conditioner shall not exceed the values specified in table I when tested in accordance with MIL-A-38339 and as specified herein.

*TABLE I. Maximum Noise Level Limits

Octave Band Center Frequencies (Hz)	Sound Level in Decibels (Ref: 0.0002 Dynes/CM^2)	
	Evaporator Side	Condenser Side
63	86	81
125	86	81
250	86	81
500	90	81
1,000	85	83
2,000	79	77
4,000	73	70
8,000	67	60

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3.9.1 Shelter. The air conditioner shall be mounted in an S-141( )/G shelter in accordance with MIL-S-52059 or an S-280( )/G shelter in accordance with MIL-S-55286. The ceiling of the shelter shall be covered with 3/4-inch acoustical fiberglass board.

* 3.10 Dimensions. The dimensions of the air conditioner shall be as follows:

	Inches
Width	30 ±1/16
Depth	18 ±1/16
Height	55 ±3/32

3.11 Weight. The weight of the complete air conditioner charged with refrigerant and oil shall not exceed 435 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection and testing. The inspection and testing of the air conditioner shall be in accordance with MIL-A-38339 and as specified herein.

4.2 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The air conditioner shall be preserved and packaged in accordance with method IIA of MIL-P-116.

5.1.2 Level C. The air conditioner shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The air conditioner, preserved and packaged as specified in 5.1.1, shall be packed in style B, type II, class 2 containers conforming to PPP-B-601. Skids shall be provided to facilitate machine handling. As far as practicable, containers shall be of uniform size and shape and of minimum tare and cube. Closure and strapping shall be in accordance with the appendix of the specification.

5.2.2 Level B. Level B packing shall be the same as level A, except the containers shall be style B, type III, class 2 of PPP-B-601. Skids shall be provided to facilitate machine handling.

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5.2.3 Level C. The air conditioner, preserved and packaged as specified in 5.1.2, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. The air conditioner shall be cushioned, anchored, blocked, and braced in accordance with MIL-STD-1186. The Edgewise drop test or Cornerwise drop test shall be in accordance with appendix A of MIL-STD-1186. Tests are not required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Air Conditioner A/E32C-25.

6. NOTES

6.1 Intended use. The A/E32C-25 air conditioner is intended for use in cooling, dehumidifying, heating, filtering, and circulating air for portable shelters or vans in accordance with electronic and personnel requirements. The air conditioner may be installed indoors or directly exposed to the atmosphere.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When external flexible ducts are to be provided (see 3.7) and lengths required

c. When a canvas cover is required (see 3.8)

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2)

e. Special shipment marking (see 5.4).

* 6.3 Drawings. Included in Trane Company DL A4524-5450-2 are the latest known revisions of all drawings available as applicable to the items specified. The drawings are not warranted by the Government to meet the specified performance requirements. In addition, the drawings are not warranted as being either a full set or sufficiently detailed for complete manufacturing purposes. The contractor is advised to carefully evaluate all drawings furnished for bidding purposes prior to preparation of his proposal or bid. It is the contractor's responsibility to evaluate this data and prepare his quotation on the basis of a comprehensive understanding of the task to be performed. The specification represents the minimum needs of the Government. The drawings furnished represent efforts of the Government to define in detail the requirements of

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the specification. Wherever appropriate, revisions or changes to the drawings shall be made to meet the performance requirements of the specification. However, changes to the drawings that effect the interchangeability of parts, assemblies, or subassemblies specified on the drawings will not be made without prior approval from the procuring activity. In the event, and to the extent approved, it becomes necessary to change drawings to meet the performance requirements of the specification, such costs of changes will be borne by the contractor and no equitable adjustment will be authorized. The inability of the Government to provide additional data or the incompleteness of any data furnished hereunder shall not later be construed as justification for an increase in contract price, or, as an excusable delay under the terms of the contract clause entitled Default.

* 6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

BTU/hr . . . . British thermal units per hour

cfm . . . . . cubic feet per minute

Hz. . . . . Hertz, cycles per second

kw. . . . . kilowatt

V . . . . . volt

w . . . . . watt

6.5 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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